Project Report

Title: Advanced Civil Material Analysis System

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Department: Electronics and Computer Engineering
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1. Project Overview

This project is a Python-based GUI application that analyzes different civil engineering materials such as Concrete, Steel, Brick, and Cement. The system allows users to input material properties and checks whether they fall within acceptable engineering ranges. Based on the input values, the program determines whether the material is SAFE or NOT SAFE.

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2. Problem Statement

In civil engineering, ensuring the quality and safety of construction materials is critical. Manual checking of material properties against standard values can be time-consuming and prone to errors. This project aims to develop an automated system that validates material properties and provides quick and accurate safety evaluation.

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3. Technology Stack

Programming Language: Python 3.x
Library Used: Tkinter (for GUI)
IDE: Python IDLE / VS Code

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4. Implementation

The implementation follows these steps:

Create a graphical user interface using Tkinter.
Provide a dropdown menu to select the material type (Concrete, Steel, Brick, Cement).
Dynamically generate input fields based on the selected material.
Define acceptable ranges for each material property.
Accept user input for each property.
Validate the input values against predefined ranges.

Display the result as:
SAFE if all values are within range
NOT SAFE if any value is out of range

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5. Code Snippet

```
# Check material safety

for key, entry in entries.items():
    value = float(entry.get())
    min_val, max_val = ranges[key]

    if value < min_val or value > max_val:
        print("Material is NOT SAFE")
        break
else:
    print("Material is SAFE")
```

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6. Sample Test Cases

Material	Input Values	Output
Concrete	Within range	SAFE
Steel	One value out of range	NOT SAFE
Brick	All values valid	SAFE
Cement	Invalid input	Error message

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7. Challenges Faced

Designing dynamic input fields based on selected material
Managing multiple property ranges efficiently
Handling invalid or non-numeric user inputs
Ensuring user-friendly interface design

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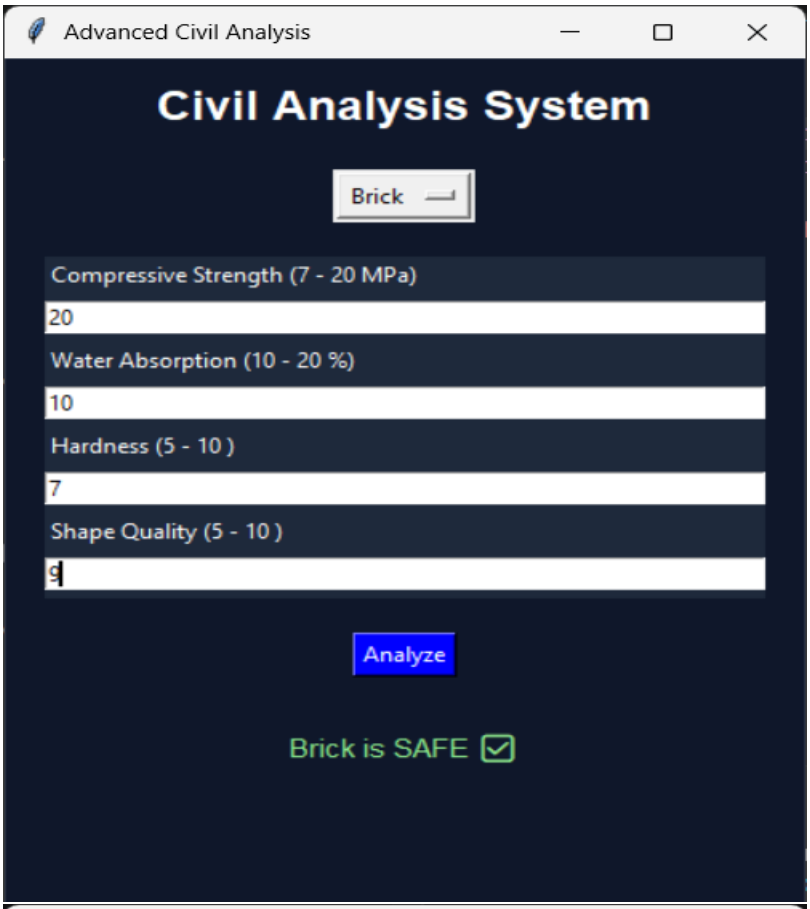
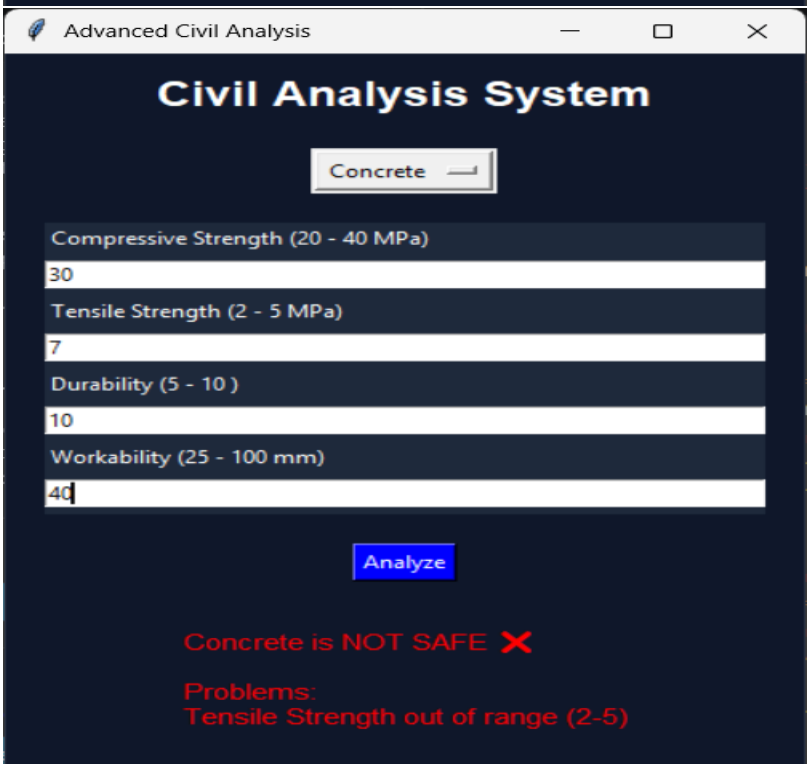
8. Results and Observations

The system successfully evaluates different materials and accurately determines their safety status. It provides immediate feedback and reduces manual calculation errors. The GUI makes it easy for users to interact with the system.


Advanced Civil Analysis
Civil Analysis System
Brick
Compressive Strength (7 - 20 MPa)
20
Water Absorption (10 - 20 %)
10
Hardness (5 - 10)
7
Shape Quality (5 - 10)
9
Analyze
Brick is SAFE


Advanced Civil Analysis
Civil Analysis System
Concrete
Compressive Strength (20 - 40 MPa)
30
Tensile Strength (2 - 5 MPa)
7
Durability (5 - 10)
10
Workability (25 - 100 mm)
40
Analyze
Concrete is NOT SAFE
Problems:
Tensile Strength out of range (2-5)

Civil Analysis System

Steel

Tensile Strength (400 - 600 MPa)

450

Yield Strength (250 - 500 MPa)

300

Ductility (10 - 25 %)

20

Hardness (120 - 200 HB)

150

Analyze

Steel is SAFE ☑

Civil Analysis System

Cement

Compressive Strength (20 - 50 MPa)

35

Setting Time (30 - 600 min)

550

Fineness (225 - 400)

450

Soundness (0 - 10 mm)

7

Analyze

Cement is NOT SAFE ✕

Problems:
Fineness out of range (225-400)

Project Submission

Repository Link: https://github.com/akshayaganapa28-prog/project

9. Signatures

Student Signature: _____

Guide Signature: _____

Date: _____